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# **Partie 1 - Conception (MCD)**

Logical database model documentation

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# 1. Model details

**Model name:**

Partie 1 - Conception (MCD)

**Version:**

1.0

**Description:**

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## 2. Entities

### 2.1. Entity Cours

**Description:**

Cours identifié par son Intitulé.

#### 2.1.1. Attributes

| Attribute name      | Type       | Is mandatory | Is primary identifier | Description |
|---------------------|------------|--------------|-----------------------|-------------|
| Id_Cours            | Integer    | yes          | yes                   | PK          |
| Intitule            | NChar(50)  | yes          | no                    |             |
| Nombre_de_parties   | Integer    | yes          | no                    |             |
| Nombre_de_chapitres | Integer    | yes          | no                    |             |
| Description         | Text(500)  | yes          | no                    |             |
| Pre_requis          | NChar(150) | no           | no                    |             |
| Prix                | Integer    | yes          | no                    |             |
| Date_de_debut       | Date       | no           | no                    |             |
| Date_de_fin         | Date       | no           | no                    |             |
| Modalite_de_suivi   | NChar(10)  | yes          | no                    |             |

### 2.2. Entity Examens

**Description:**

Examen portant sur certaines parties d'un cours, identifié par son titre.

#### 2.2.1. Attributes

| Attribute name             | Type       | Is mandatory | Is primary identifier | Description |
|----------------------------|------------|--------------|-----------------------|-------------|
| Id_Examen                  | Integer    | yes          | yes                   | PK          |
| Titre                      | NChar(30)  | yes          | no                    |             |
| Partie_numero_tee_du_cours | Integer    | yes          | no                    | FK          |
| Contenu_textuel            | Text(1500) | yes          | no                    |             |
| Score_minimum              | Integer    | yes          | no                    |             |

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## 2.3. Entity Partie\_numerotee\_cours

### Description:

Partie unique d'un cours associée à un examen, identifié par son Id\_Partie.

#### 2.3.1. Attributes

| Attribute name  | Type       | Is mandatory | Is primary identifier | Description |
|-----------------|------------|--------------|-----------------------|-------------|
| Id_Partie       | Integer    | yes          | yes                   | PK          |
| Id_Cours        | Integer    | yes          | no                    | FK          |
| Titre           | NChar(50)  | yes          | no                    |             |
| Contenu         | Text(1500) | yes          | no                    |             |
| Chapitre_numero | Integer    | no           | no                    |             |

## 2.4. Entity Sessions

### Description:

Session de cours (en autonomie, en direct présentiel ou en direct distanciel)

#### 2.4.1. Attributes

| Attribute name           | Type      | Is mandatory | Is primary identifier | Description |
|--------------------------|-----------|--------------|-----------------------|-------------|
| Id_Session               | Integer   | yes          | yes                   | PK          |
| Id_Formateur             | Integer   | yes          | no                    | FK          |
| Id_Cours                 | Integer   | yes          | no                    | FK          |
| Date_de_debut            | Date      | yes          | no                    |             |
| Date_de_fin              | Date      | yes          | no                    |             |
| Heure_de_debut           | Time      | yes          | no                    |             |
| Heure_de_fin             | Time      | yes          | no                    |             |
| Nombre_de_places_maximum | Integer   | yes          | no                    |             |
| Modalites_de_suivi       | NChar(10) | yes          | no                    |             |

## 2.5. Entity Etudiants

### Description:

Correspond à l'ensemble des étudiants inscrits, identifiés par leur Id\_Etudiant.

### 2.5.1. Attributes

| Attribute name   | Type    | Is mandatory | Is primary identifier | Description |
|------------------|---------|--------------|-----------------------|-------------|
| Id_Etudiant      | Integer | yes          | yes                   | PK          |
| Date_Inscription | Date    | yes          | no                    |             |

## 2.6. Entity Formateurs

### Description:

Correspond à l'ensemble des utilisateurs hors étudiants, identifiés par leur Id\_Utilisateur.

### 2.6.1. Attributes

| Attribute name          | Type      | Is mandatory | Is primary identifier | Description |
|-------------------------|-----------|--------------|-----------------------|-------------|
| Id_Utilisateur          | Integer   | yes          | yes                   | PK          |
| Nom                     | NChar(30) | yes          | no                    |             |
| Nature_de_l_Utilisateur | Integer   | yes          | no                    |             |

## 2.7. Entity Tentatives

### Description:

Correspond à une tentative de passage d'un examen par un étudiant.

### 2.7.1. Attributes

| Attribute name     | Type           | Is mandatory | Is primary identifier | Description |
|--------------------|----------------|--------------|-----------------------|-------------|
| Id_Tentative       | Integer        | yes          | yes                   | PK          |
| Id_Etudiant        | Integer        | yes          | no                    | FK          |
| Id_Examen          | Integer        | yes          | no                    | FK          |
| Score              | Integer        | yes          | no                    |             |
| Date               | Date           | yes          | no                    |             |
| Id_Formateur       | Integer        | yes          | no                    | FK          |
| Note               | Decimal(20, 2) | yes          | no                    |             |
| Statut_de_reussite | NChar(11)      | yes          | no                    |             |

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## 2.8. Entity Appreciations\_cours

### Description:

Table intermédiaire permettant d'associer l'appréciation d'un étudiant à un cours.

#### 2.8.1. Attributes

| Attribute name | Type       | Is mandatory | Is primary identifier | Description |
|----------------|------------|--------------|-----------------------|-------------|
| Id_Cours       | Integer    | yes          | yes                   | PK, FK      |
| Id_Etudiant    | Integer    | yes          | yes                   | PK, FK      |
| Note           | Integer    | yes          | no                    |             |
| Commentaire    | NChar(100) | no           | no                    |             |

## 2.9. Entity Inscription

### Description:

Table intermédiaire permettant d'associer un étudiant inscrit à un cours.

#### 2.9.1. Attributes

| Attribute name | Type    | Is mandatory | Is primary identifier | Description |
|----------------|---------|--------------|-----------------------|-------------|
| Id_Cours       | Integer | yes          | yes                   | PK, FK      |
| Id_Etudiant    | Integer | yes          | yes                   | PK, FK      |

## 2.10. Entity Formateur\_Cours

### Description:

Table intermédiaire permettant d'associer un formateur à un cours.

#### 2.10.1. Attributes

| Attribute name | Type    | Is mandatory | Is primary identifier | Description |
|----------------|---------|--------------|-----------------------|-------------|
| Id_Cours       | Integer | yes          | yes                   | PK, FK      |
| Id_Formateur   | Integer | yes          | yes                   | PK, FK      |

## 2.11. Entity Formateur\_Session

### Description:

Table intermédiaire permettant d'associer un formateur à une session.

#### 2.11.1. Attributes

| Attribute name | Type    | Is mandatory | Is primary identifier | Description |
|----------------|---------|--------------|-----------------------|-------------|
| Id_Session     | Integer | yes          | yes                   | PK, FK      |
| Id_Formateur   | Integer | yes          | yes                   | PK, FK      |

## 2.12. Entity Parties\_validees

### Description:

Table intermédiaire permettant d'associer la validation d'un cours d'un étudiant.

### 2.12.1. Attributes

| Attribute name | Type    | Is mandatory | Is primary identifier | Description |
|----------------|---------|--------------|-----------------------|-------------|
| Id_Etudiant    | Integer | yes          | yes                   | PK, FK      |
| Id_Partie      | Integer | yes          | yes                   | PK, FK      |

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## 3. Associations

### 3.1. Association Redige

| Attribute name | Type | Is mandatory | Description |
|----------------|------|--------------|-------------|
|----------------|------|--------------|-------------|

### 3.2. Association S\_inscrit

| Attribute name | Type | Is mandatory | Description |
|----------------|------|--------------|-------------|
|----------------|------|--------------|-------------|

### 3.3. Association Concerne

| Attribute name | Type | Is mandatory | Description |
|----------------|------|--------------|-------------|
|----------------|------|--------------|-------------|

### 3.4. Association Est\_compose\_de

| Attribute name | Type | Is mandatory | Description |
|----------------|------|--------------|-------------|
|----------------|------|--------------|-------------|

### 3.5. Association Est\_parmi

| Attribute name | Type | Is mandatory | Description |
|----------------|------|--------------|-------------|
|----------------|------|--------------|-------------|

### 3.6. Association Supervise

| Attribute name | Type | Is mandatory | Description |
|----------------|------|--------------|-------------|
|----------------|------|--------------|-------------|

### 3.7. Association Est\_un

| Attribute name | Type | Is mandatory | Description |
|----------------|------|--------------|-------------|
|----------------|------|--------------|-------------|

### 3.8. Association Est\_attribue\_a

| Attribute name | Type | Is mandatory | Description |
|----------------|------|--------------|-------------|
|----------------|------|--------------|-------------|

### 3.9. Association Realise

| Attribute name | Type | Is mandatory | Description |
|----------------|------|--------------|-------------|
|----------------|------|--------------|-------------|

### 3.10. Association Corrige

| Attribute name | Type | Is mandatory | Description |
|----------------|------|--------------|-------------|
|----------------|------|--------------|-------------|

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### 3.11. Association D\_une

| Attribute name | Type | Is mandatory | Description |
|----------------|------|--------------|-------------|
|----------------|------|--------------|-------------|

### 3.12. Association D\_un

| Attribute name | Type | Is mandatory | Description |
|----------------|------|--------------|-------------|
|----------------|------|--------------|-------------|

### 3.13. Association A\_un

| Attribute name | Type | Is mandatory | Description |
|----------------|------|--------------|-------------|
|----------------|------|--------------|-------------|

### 3.14. Association Assure

| Attribute name | Type | Is mandatory | Description |
|----------------|------|--------------|-------------|
|----------------|------|--------------|-------------|

### 3.15. Association A\_valide

| Attribute name | Type | Is mandatory | Description |
|----------------|------|--------------|-------------|
|----------------|------|--------------|-------------|

### 3.16. Association Correspond\_a

| Attribute name | Type | Is mandatory | Description |
|----------------|------|--------------|-------------|
|----------------|------|--------------|-------------|

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## 4. Entity relationships

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## 5. Association relationships

### 5.1. Association relationship Association\_29

|        |     |           |
|--------|-----|-----------|
| Redige |     | Etudiants |
|        | <-> |           |

### 5.2. Association relationship Association\_1

|        |     |                     |
|--------|-----|---------------------|
| Redige |     | Appreciations_cours |
|        | <-> |                     |

### 5.3. Association relationship Association\_32

|           |     |           |
|-----------|-----|-----------|
| S_inscrit |     | Etudiants |
|           | <-> |           |

### 5.4. Association relationship Association\_17

|           |     |             |
|-----------|-----|-------------|
| S_inscrit |     | Inscription |
|           | <-> |             |

### 5.5. Association relationship Association\_31

|         |     |           |
|---------|-----|-----------|
| Realise |     | Etudiants |
|         | <-> |           |

### 5.6. Association relationship Association\_28

|         |     |            |
|---------|-----|------------|
| Realise |     | Tentatives |
|         | <-> |            |

### 5.7. Association relationship Association\_26

|         |     |            |
|---------|-----|------------|
| Corrige |     | Tentatives |
|         | <-> |            |

### 5.8. Association relationship Association\_14

|         |     |            |
|---------|-----|------------|
| Corrige |     | Formateurs |
|         | <-> |            |

### 5.9. Association relationship Association\_27

|      |     |            |
|------|-----|------------|
| A_un |     | Tentatives |
|      | <-> |            |

### 5.10. Association relationship Association\_9

|      |     |         |
|------|-----|---------|
| A_un |     | Examens |
|      | <-> |         |

### 5.11. Association relationship Association\_8

|       |     |         |
|-------|-----|---------|
| D_une |     | Examens |
|       | <-> |         |

### 5.12. Association relationship Association\_19

|       |     |                        |
|-------|-----|------------------------|
| D_une |     | Partie_numerotee_cours |
|       | <-> |                        |

### 5.13. Association relationship Association\_5

|                |     |       |
|----------------|-----|-------|
| Est_compose_de |     | Cours |
|                | <-> |       |

### 5.14. Association relationship Association\_20

|                |     |                        |
|----------------|-----|------------------------|
| Est_compose_de |     | Partie_numerotee_cours |
|                | <-> |                        |

### 5.15. Association relationship Association\_3

|      |     |       |
|------|-----|-------|
| D_un |     | Cours |
|      | <-> |       |

### 5.16. Association relationship Association\_24

|      |      |          |
|------|------|----------|
| D_un |      | Sessions |
|      | <--> |          |

### 5.17. Association relationship Association\_25

|           |      |          |
|-----------|------|----------|
| Supervise |      | Sessions |
|           | <--> |          |

### 5.18. Association relationship Association\_13

|           |      |                   |
|-----------|------|-------------------|
| Supervise |      | Formateur_Session |
|           | <--> |                   |

### 5.19. Association relationship Association\_16

|        |      |            |
|--------|------|------------|
| Est_un |      | Formateurs |
|        | <--> |            |

### 5.20. Association relationship Association\_12

|        |      |                   |
|--------|------|-------------------|
| Est_un |      | Formateur_Session |
|        | <--> |                   |

### 5.21. Association relationship Association\_18

|          |      |             |
|----------|------|-------------|
| Concerne |      | Inscription |
|          | <--> |             |

### 5.22. Association relationship Association\_4

|          |      |       |
|----------|------|-------|
| Concerne |      | Cours |
|          | <--> |       |

### 5.23. Association relationship Association\_6

|                |      |       |
|----------------|------|-------|
| Est_attribue_a |      | Cours |
|                | <--> |       |

## 5.24. Association relationship Association\_2

|                |     |                     |
|----------------|-----|---------------------|
| Est_attribue_a |     | Appreciations_cours |
|                | <-> |                     |

## 5.25. Association relationship Association\_10

|           |     |                 |
|-----------|-----|-----------------|
| Est_parmi |     | Formateur_Cours |
|           | <-> |                 |

## 5.26. Association relationship Association\_15

|           |     |            |
|-----------|-----|------------|
| Est_parmi |     | Formateurs |
|           | <-> |            |

## 5.27. Association relationship Association\_11

|        |     |                 |
|--------|-----|-----------------|
| Assure |     | Formateur_Cours |
|        | <-> |                 |

## 5.28. Association relationship Association\_7

|        |     |       |
|--------|-----|-------|
| Assure |     | Cours |
|        | <-> |       |

## 5.29. Association relationship Association\_30

|          |     |           |
|----------|-----|-----------|
| A_valide |     | Etudiants |
|          | <-> |           |

## 5.30. Association relationship Association\_22

|          |     |                  |
|----------|-----|------------------|
| A_valide |     | Parties_validees |
|          | <-> |                  |

## 5.31. Association relationship Association\_23

|              |     |                  |
|--------------|-----|------------------|
| Correspond_a |     | Parties_validees |
|              | <-> |                  |

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## 5.32. Association relationship Association\_21

|              |     |                        |
|--------------|-----|------------------------|
| Correspond_a |     | Partie_numerotee_cours |
|              | <-> |                        |

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## 6. Areas

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## 7. Notes